

The stated coercivity assumption cannot hold on a nonempty domain. Indeed, setting $v = 0$ gives $0 = a(u, 0) \geq c\|u\|_{1,\Omega}^2$ for every $u \in H_0^1(\Omega)$. Since $c > 0$, this would make $H_0^1(\Omega) = \{0\}$. But a nonempty domain contains a nonzero function in $C_0^\infty(\Omega) \subset H_0^1(\Omega)$. Thus the assumptions as written are inconsistent, and the asserted result has no nonvacuous proof under them.

With the intended, diagonal coercivity assumption $a(u, u) \geq c\|u\|_{1,\Omega}^2$ for every $u \in H_0^1(\Omega)$, the proof is as follows. The space $H_0^1(\Omega)$ is a Hilbert space, a is bounded and coercive on it, and $F \in H^{-1}(\Omega)$ is continuous. The Lax–Milgram Lemma therefore gives a unique $y \in H_0^1(\Omega)$ satisfying $a(y, v) = F(v)$ for all $v \in H_0^1(\Omega)$. The finite-dimensional space V_h^0 , equipped with the inherited norm, is also a Hilbert space. The restrictions of a and F to this space satisfy the same boundedness and coercivity conditions, so the Lax–Milgram Lemma gives a unique $y_h \in V_h^0$ satisfying $a(y_h, v_h) = F(v_h)$ for all $v_h \in V_h^0$.

Set $e = y - y_h$. Subtracting the variational equations yields Galerkin orthogonality: $a(e, w_h) = 0$ for every $w_h \in V_h^0$. For any $v_h \in V_h^0$, we have $v_h - y_h \in V_h^0$, and hence $a(e, e) = a(e, y - v_h)$. Coercivity and boundedness now give

$$c\|e\|_{1,\Omega}^2 \leq a(e, e) \leq C\|e\|_{1,\Omega}\|y - v_h\|_{1,\Omega}.$$

If $e \neq 0$, cancellation followed by taking the infimum over $v_h \in V_h^0$ proves

$$\|y - y_h\|_{1,\Omega} \leq (C/c) \inf_{v_h \in V_h^0} \|y - v_h\|_{1,\Omega}.$$

If $e = 0$, the estimate is immediate.